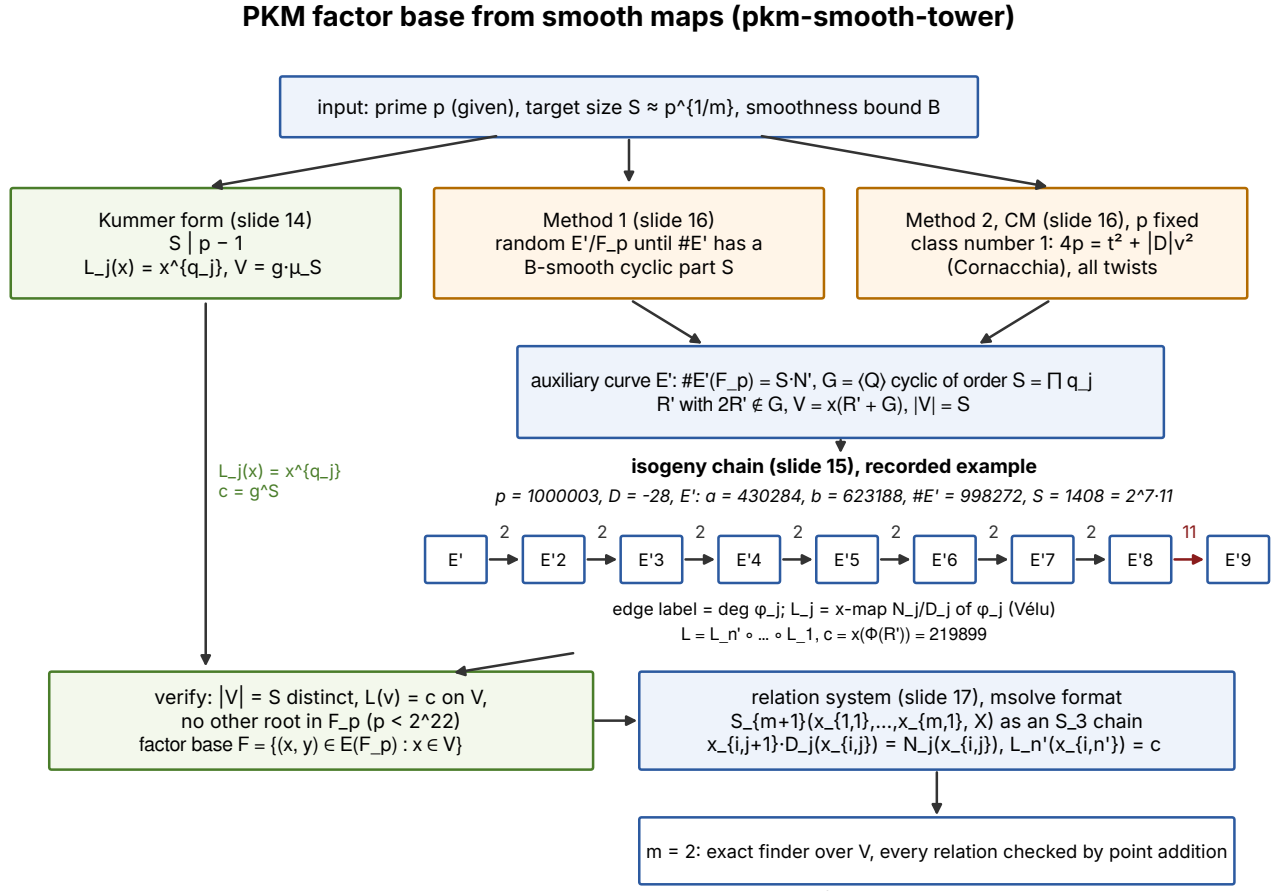
1. PKM factor bases from smooth maps: `pkm-smooth-tower`

2026-11-14 · isogeny-labs/crypto PR #4091 · code at PR head · status: construction and checking tooling; tests pass, nothing benchmarked [pcc:unverified].

1.1. What was built

Petit–Kosters–Messeng (PKC 2016, slides 12–17) take a factor base $\mathcal{F} = \{(x, y) \in E(\mathbb{F}_p) : x \in V\}$ with $V = \{x : L(x) = c\}$, where $L = L_{n'} \circ \cdots \circ L_1$ is a composition of low-degree maps and $|V| \approx \prod \deg L_j \approx p^{1/m}$. Until now the repository built V from degree-2 maps only (`ic_framework::prime_tower`). The module `src/cryptanalysis/pkm_smooth_tower.rs` and the CLI `pkm-smooth-tower` accept any ordered list of small primes $q_1, \ldots, q_{n'}$, with $S = \prod q_j$:

- **Kummer form** (slide 14): $L_j(x) = x^{q_j}$ when $S \mid p - 1$; $V = g \cdot \mu_S$.
- **Isogeny form** (slide 15): an auxiliary E' with cyclic $G = \langle Q \rangle$ of order S. Each L_j is the x-map N_j/D_j of a Vélu isogeny of degree q_j, of either parity. $V = x(R' + G)$ with $2R' \notin G$, and $c = x(\Phi(R'))$.
- **Method 1** (slide 16): random curves until $\#E'$ has a B-smooth cyclic part S; the number of curves tried is recorded.
- **Method 2, CM** (slide 16), for the **given** p: the 13 class-number-1 orders, Cornacchia for $4p = t^2 + |D|\ v^2$, every twist, orders certified on random points; and with class number > 1: every $|D| \le M$ with $h(D) \le H$, the class polynomial H_D (`isogeny_algos::cm`) reduced mod p, its roots by Cantor–Zassenhaus, every twist certified as $p + 1 \mp t$. Choosing p itself ("kind of trapdoor") is out of scope.
- **Checks**: $|V| = S$ distinct values, $L(v) = c$ on V, and below 2^{22} no other root in $\mathbb{F}_p$. **System** (slide 17): msolve export with S_{m+1} written as an S_3 chain; an exact $m = 2$ finder checks every relation by point addition.



Source: research/pkm_smooth_tower_20261114/data/cm_p1000003.json (pkm-smooth-tower at a894bbdaa9). Red arrows: odd-degree steps.

Figure 1: The construction as implemented, with the recorded CM example (`data/cm_p1000003.json`): a chain of seven 2-isogenies and one 11-isogeny on a $D = -28$ curve, $S = 1408$.

1.2. Recorded examples (measured on this container, no isolation)

command	route	result	record
`kummer --p 20071 --degrees 3,2,5,3`	Kummer	$S = 90$, verified	`data/kummer_p20071.json`

`isogeny --p 1000003 --bound 13 --min-size 500 --seed 2`	Method 1	$S = 720 = 2^4 \cdot 3^2 \cdot 5$, 39 curves tried, verified	`data/isogeny_p1000003.json`
`cm --p 1000003 --bound 13 --min-size 300`	Method 2	$D = -28$, $S = 1408 = 2^7 \cdot 11$, verified	`data/cm_p1000003.json`
`search --p 1000003 --bound 13 --min-size 500`, seeds 1–20	Method 1	curves tried: mean 17.4, min 1, max 68	`data/method1_p1000003_seeds1-20.json`

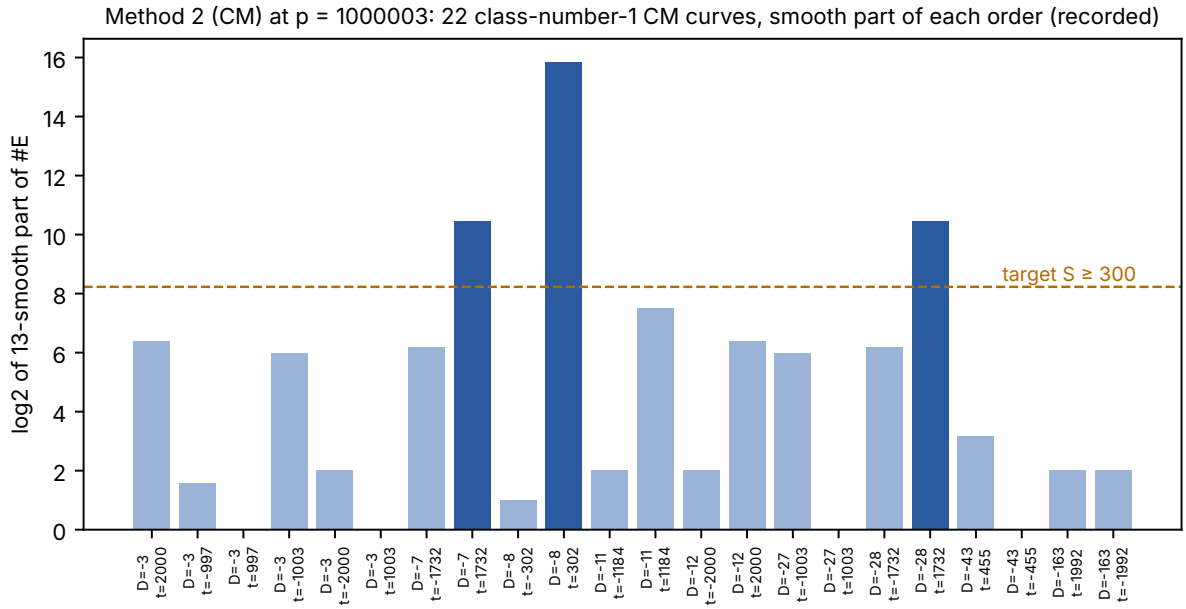


Figure 2: Method 2 at $p = 1000003$: the 22 class-number-1 CM curves and the 13-smooth part of each order (recorded). Three reach $S \geq 300$. `cm_aux` returned the $D = -28$ curve (order 998272). The $D = -7$ curve of the same order and the $D = -8$ curve with the largest smooth part were passed over. `cm_aux` takes the first fit, not the largest, and why it skipped $D = -7$ is not established here.

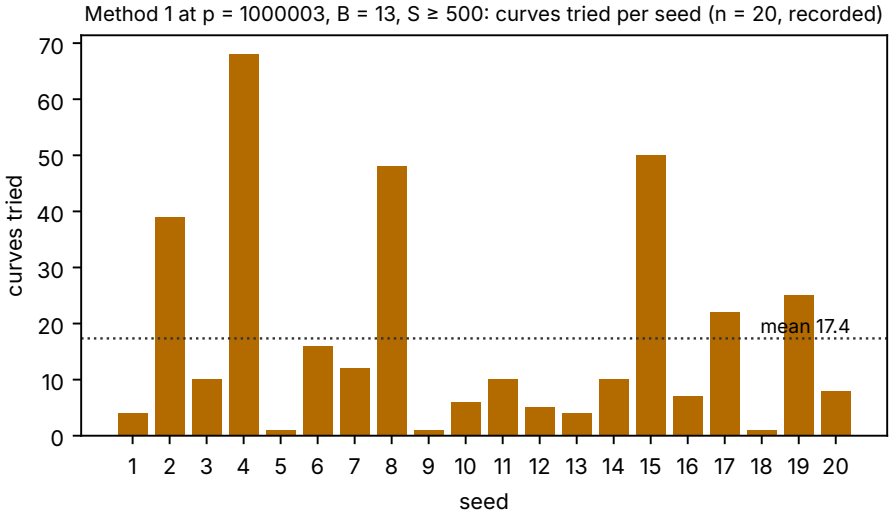


Figure 3: Method 1 at $p = 1000003$, $B = 13$, $S \geq 500$: curves tried per seed, $n = 20$ (recorded).

1.3. Class number > 1 (Hilbert class polynomials)

`cm --p P --bound 13 --min-size 300 --max-disc 1000 --max-h 16`, recorded in `data/cm_hilbert_p*_D1000_h16.json`; wall time 1.1–1.5 s each on this container (measured once, unisolated).

p	D scanned / split	roots	rejected / incomplete	CM curves	tower
1000003	452 / 54	262	0 / 0	528	$D = -28$, $S = 1408 = 2^7 \cdot 11$ (as class number 1)

1000033	452 / 78	388	0 / 0	782	$D = -99$, $S = 891 = 3^4 \cdot 11$
1000039	452 / 48	263	0 / 0	530	$D = -124$, $S = 800 = 2^5 \cdot 5^2$

At $p = 1000033$ and 1000039 no class-number-1 curve fits $S \geq 300$; the larger scan finds one at each. Every split D gave $h(D)$ roots and every root a certified curve, which is the check that the rounded H_D is right for that use.

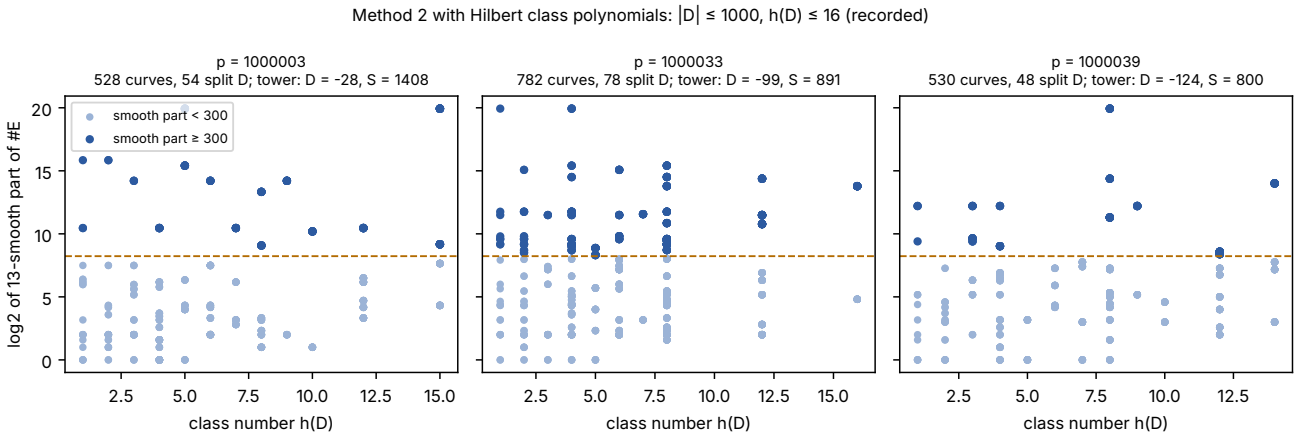


Figure 4: Smooth part of every CM curve order against $h(D)$ at three primes (recorded).

1.4. Checks

`cargo test --release --lib pkm_smooth_tower`: 8 passed. The tests cover:

- Vélu x-maps agreeing with the point map, and codomain orders equal to domain orders.
- Mixed Kummer and mixed isogeny towers.
- Real $m = 2$ decompositions.
- CM orders matching the generic point count, with the expected number of twists per D, at 4 primes up to 998244353.
- A CM curve feeding a verified tower.
- Root finding mod p against planted roots (with an irreducible quadratic factor).
- The Hilbert scan at two primes: no failed polynomial, no incomplete split, no rejected root. Its orders for $h > 1$ match the generic count, and it reproduces every class-number-1 curve.

1.5. Scope and next steps

Single-word arithmetic ($p < 2^{62}$); order counting is practical to about 48 bits. No ecbench plug-in, so nothing is priced against the index-calculus baseline or rho. The class polynomial is computed over $\mathbb{C}$ in fixed point, and has been run here only to $|D| \leq 1000$, $h \leq 16$. `cm_aux_from` is first-fit, not best-fit. Next: an `ic_framework` factor-base plug-in so the mixed tower can be charged in ecbench's unit.